

L8: Dynamic Communities – Approach #3

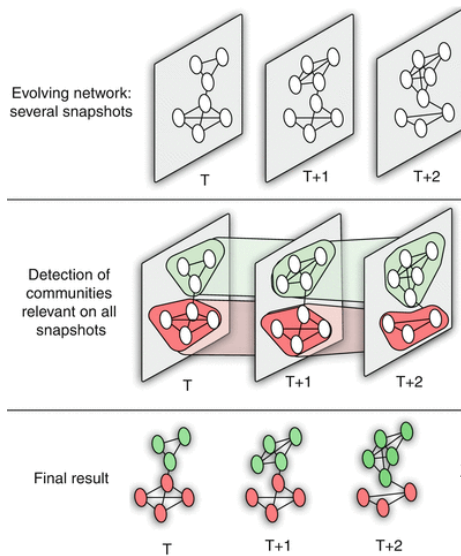


Figure 4 from [Dynamic Community Detection](https://doi.org/10.1007/978-1-4614-6170-8_383) (https://doi.org/10.1007/978-1-4614-6170-8_383) by Cazabet R., Amblard F.

A third approach is to consider all network snapshots simultaneously. Or, a more pragmatic approach would be to consider an entire “**window**” of network snapshots ($T, T + 1, T + 2, \dots, T + W - 1$) at the same time,

Algorithms of this type create a single “**global network**” based on all these snapshots as follows:

1. A node of even a single snapshot is also a node of the global network, and an edge of even a single snapshot is also an edge between the corresponding two nodes at the global network.
2. Additionally, the global network includes edges between the instances of any node X at different snapshots (so if X appears at time T and at time $T+1$, there will be an edge between the corresponding two node instances in the global network).
3. After creating the global network, a community detection algorithm is applied on that

network to compute a “**reference**” community structure.

4. Then, that reference community structure is mapped back to each snapshot, based on the nodes and edges that are present at that snapshot.

This approach usually produces the smoothest (or *most stable*) results – but it can miss major steps in the evolution of the community structure (such as communities that merge or split). This last point however also depends on the length of the window W .

Food for Thought

Explain in more detail the rationale for constructing the “global network” and for computing a “reference” community structure based on that network.